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ORIGINAL ARTICLE

Subcutaneous versus intravenous pembrolizumab, in combination with chemotherapy, for treatment of metastatic non-small-cell lung cancer: the phase III 3475A-D77 trial

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Background: Pembrolizumab with berahyaluronidase alfa is for subcutaneous (s.c.) administration. The phase III open-label 3475A-D77 study (NCT05722015) assessed s.c. pembrolizumab versus intravenous (i.v.) pembrolizumab, plus chemotherapy, for treatment of metastatic non-small-cell lung cancer (mNSCLC).

Patients and methods: Participants with newly diagnosed stage IV squamous or nonsquamous NSCLC without sensitizing *EGFR*, *ALK*, or *ROS1* alterations were randomized 2 : 1 to pembrolizumab s.c. 790 mg every 6 weeks (q6w) or pembrolizumab i.v. 400 mg q6w (18 cycles), each given with platinum-doublet chemotherapy. Dual primary endpoints were pharmacokinetic exposure measures of cycle 1 area under the curve ($AUC_{0\text{-}6\ weeks}$) and steady-state trough concentration (C_{trough}) of pembrolizumab. The noninferiority margin for $AUC_{0\text{-}6\ weeks}$ and C_{trough} geometric mean ratios (GMRs) of pembrolizumab s.c. versus i.v. was specified as 0.8. Secondary endpoints included additional pharmacokinetic exposure measures, pembrolizumab immunogenicity, efficacy, and safety.

Results: In total 377 participants were randomized to the pembrolizumab s.c. ($n = 251$) or i.v. ($n = 126$) arms. The median time from randomization to data cut-off (12 July 2024) was 9.6 months (range 6.2-16.4 months). The median injection time for pembrolizumab s.c. was 2.0 min (range 1-12 min). The GMR [96% confidence interval (CI)] for cycle 1 $AUC_{0\text{-}6\ weeks}$ was 1.14 (1.06-1.22); $P < 0.0001$. The GMR (94% CI) for steady-state C_{trough} was 1.67 (1.52-1.84); $P < 0.0001$. Secondary pharmacokinetic endpoints were within established bounds for pembrolizumab. Anti-pembrolizumab antibodies were detected in 1.4% (pembrolizumab s.c. arm) and 0.9% (pembrolizumab i.v. arm) of participants. For the pembrolizumab s.c. versus i.v. arms, objective response rates (ORRs) were 45.4% versus 42.1% (ORR ratio 1.08, 95% CI 0.85-1.37). Other efficacy measures were similar and safety profiles were consistent between treatment arms.

Conclusions: Overall exposure and trough concentrations of pembrolizumab s.c. 790 mg q6w were noninferior to those of pembrolizumab i.v. 400 mg q6w given with chemotherapy in participants with treatment-naive mNSCLC. Results support pembrolizumab s.c. as a treatment option in all indications where pembrolizumab i.v. can be used.

Key words: immunotherapy, non-small-cell lung cancer, pembrolizumab, phase III, programmed cell death protein 1, subcutaneous

INTRODUCTION

Subcutaneous (s.c.) administration of therapeutic agents has been shown to have advantages over intravenous (i.v.) administration, including higher patient satisfaction, lower emotional burden (e.g. less anxiety and stress), and greater patient and provider preference.[1-4] Moreover, s.c.

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administration has been associated with improvements in administration workflows, including shorter clinic time and provider time.[5-10]

Pembrolizumab administered by i.v. infusion is approved globally for the treatment of multiple tumor types, including non-small-cell lung cancer (NSCLC). Subcutaneously administered pembrolizumab would have the potential to increase convenience for patients and providers, increase patient satisfaction, and improve clinic efficiency. MK-3475A is pembrolizumab with berahyaluronidase alfa (MK-5180), a variant of human hyaluronidase developed and manufactured by Alteogen Inc. (Daejeon, South Korea), for s.c. administration. Berahyaluronidase alfa acts as a permeation enhancer that temporarily degrades hyaluronan in the extracellular matrix surrounding the injection area, increasing dispersion and enabling s.c. administration of both every 3 weeks (q3w) and every 6 weeks (q6w) dosing regimens of pembrolizumab. The phase I 3475A-C18 study (NCT05017012) assessed the bioavailability, pharmacokinetics, safety, and tolerability of s.c. pembrolizumab ('pembrolizumab s.c.') in participants with advanced or metastatic melanoma, renal cell carcinoma (RCC), and NSCLC. Results of the C18 study informed the selection of the pembrolizumab dose and the pembrolizumab s.c. solution strength,[11,12] pembrolizumab s.c. 790 mg at 165 mg/ml q6w, for further clinical development.

Here, we report the findings of the phase III, randomized 3475A-D77 study of pembrolizumab s.c. versus pembrolizumab i.v., in combination with chemotherapy, for the treatment of participants with newly diagnosed metastatic NSCLC (mNSCLC).

PATIENTS AND METHODS

Study design and treatments

In the phase III, global, open-label 3475A-D77 study (NCT05722015), participants at 61 sites (Supplementary Table S1, available at https://doi.org/10.1016/j.annonc.2025.03.012) were randomized 2 : 1 to pembrolizumab s.c. or pembrolizumab i.v., administered in combination with platinum-doublet chemotherapy. Randomization was stratified by Eastern Cooperative Oncology Group (ECOG) performance status (0 versus 1), histology (squamous versus nonsquamous), programmed death-ligand 1 (PD-L1) tumor proportion score [$<$50% versus $\geq$50% as assessed centrally as per PD-L1 IHC 22C3 pharmDx (Agilent Technologies, Carpinteria, CA)], and geographic region (east Asia versus North America/western Europe/Australia/New Zealand versus rest of the world).

Pembrolizumab s.c. 790 mg was administered subcutaneously in the abdominal area or thigh q6w at a solution strength of 165 mg/ml with berahyaluronidase alfa 2000 U/ml. The total injection volume was $\sim$4.8 ml. Pembrolizumab i.v. 400 mg was administered intravenously q6w as an infusion over $\sim$30 min. In both arms, pembrolizumab was administered for up to 18 cycles. Participants with nonsquamous NSCLC received up to four infusions of pemetrexed 500 mg/m^2 q3w with investigator's choice of cisplatin 75 mg/m^2 q3w or carboplatin area under the concentration—time curve (AUC) 5 mg/ml/min q3w followed by pemetrexed maintenance. Participants with squamous NSCLC received up to four infusions of carboplatin AUC 5 mg/ml/min q3w with investigator's choice of paclitaxel 200 mg/m^2 q3w or nab-paclitaxel 100 mg/m^2 on days 1, 8, 15, 22, 29, and 36 of cycles 1 and 2. This trial was conducted in compliance with global standards, local and/or national regulations of each study site, and International Council for Harmonisation Good Clinical Practice guidelines, and in accordance with the ethical principles that have their origin in the Declaration of Helsinki.

Study participants

Eligible participants were adults $\geq$18 years of age who were newly diagnosed with stage IV, histologically or cytologically confirmed squamous or nonsquamous NSCLC with no tumor-activating *EGFR* mutations or *ALK* or *ROS1* gene rearrangements. All participants had measurable disease according to RECIST, version 1.1 (RECIST v1.1)[13] and an ECOG performance status score of 0 or 1. All participants provided written informed consent. Full eligibility criteria are listed in Supplementary Table S2, available at https://doi.org/10.1016/j.annonc.2025.03.012.

Endpoints and assessments

The dual primary endpoints were cycle 1 AUC from week 0 to week 6 ($AUC_{0\text{-}6\ weeks}$) and steady-state (cycle 3) trough concentrations (C_{trough}) of pembrolizumab. Secondary pharmacokinetics endpoints were cycle 1 peak concentration (C_{max}) and C_{trough}, steady-state (cycle 3) $AUC_{0\text{-}6\ weeks}$ and C_{max} of pembrolizumab, and cycle 1 and steady-state model-based C_{trough} of pembrolizumab. Model-based and observed assessments of pharmacokinetic endpoints were carried out, as feasible. Additional details about the pharmacokinetic analysis and population pharmacokinetic modeling are provided in the Supplementary Materials, available at https://doi.org/10.1016/j.annonc.2025.03.012. During cycle 1, blood samples for pharmacokinetic endpoints were collected predose and postdose on days 2-7, 10, 15, 29, and 42 in the pembrolizumab s.c. arm and predose, within 30 min postinfusion, and on days 4, 15, 29, and 42 in the pembrolizumab i.v. arm. In cycle 3, blood samples for pharmacokinetic endpoints were collected predose and postdose on days 4, 10, and 42 in the pembrolizumab s.c. arm and predose, within 30 min postinfusion, and on days 4 and 42 in the pembrolizumab i.v. arm.

To assess the presence of circulating anti-drug antibodies (ADAs) against pembrolizumab, a secondary endpoint, blood samples were collected predose and postdose on day 42 in cycles 1 and 3 and predose in cycles 6, 8, 10, 12, 14, 16, and 18 for both arms.

Secondary efficacy endpoints were objective response rate [ORR, defined as the percentage of participants with a confirmed complete or partial response (CR or PR)], progression-free survival (PFS, defined as the time from

randomization to first documented disease progression or death from any cause, whichever occurs first), and duration of response (DOR, defined as the time from first documented CR or PR until disease progression or death from any cause, whichever occurs first) based on RECIST v1.1 as assessed by blinded independent central review, as well as overall survival (OS, defined as the time from randomization to death from any cause). Tumor imaging was carried out at baseline and postbaseline at weeks 6, 12, and 18, every 9 weeks thereafter until week 45, and then every 12 weeks. Participants with known brain metastases were required to undergo brain imaging during screening to confirm the stability of the metastases, at week 6, and then if clinically indicated or to confirm CR.

The incidence and causality of adverse events (AEs) and the discontinuation of study treatment due to AEs were secondary endpoints. AEs and laboratory abnormalities were collected regularly throughout the study and for 30 days after the last dose of study treatment. Participants were followed for the occurrence of serious AEs for 90 days or until the start of new anticancer therapy, whichever was earlier.

Change from baseline to week 24 in global health status/quality of life (QoL), physical functioning, and role functioning as assessed using the European Organisation for Research and Treatment of Cancer (EORTC) Core Quality of Life Questionnaire (QLQ-C30) were secondary endpoints. Change from baseline to week 24 in the EuroQol 5-dimension, five-level questionnaire (EQ-5D-5L) was a prespecified exploratory endpoint. Additional details for EORTC QLQ-C30 and EQ-5D-5L completion are provided in the Supplementary Materials, available at https://doi.org/10.1016/j.annonc.2025.03.012.

Statistical analysis

Pharmacokinetics were assessed in the per-protocol population, defined as participants who received the cycle 1 dose and had one or more valid postdose pharmacokinetic sample in cycle 1 and for whom a model-based assessment of cycle 1 $AUC_{0\text{-}6\ weeks}$ could be made, and participants who received the cycle 1 and cycle 3 doses and had one or more valid postdose pharmacokinetic sample in cycle 1 and for whom a model-based assessment of cycle 3 C_{trough} could be made. The dual primary hypotheses were noninferiority of pembrolizumab s.c. versus pembrolizumab i.v. with respect to the geometric mean ratios (GMRs) of cycle 1 $AUC_{0\text{-}6\ weeks}$ and cycle 3 model-based C_{trough}. The noninferiority margin for the $AUC_{0\text{-}6\ weeks}$ and C_{trough} was prespecified as 0.8. Hypothesis testing for noninferiority and the confidence intervals (CIs) for GMR were calculated using Welch's *t*-test statistics with the log-transformed $AUC_{0\text{-}6\ weeks}$ and C_{trough}. The graphical approach of Maurer et al.[14] was used to control the overall type I error between the primary endpoints at a one-sided α level of 0.05, with 0.02 initially allocated to cycle 1 $AUC_{0\text{-}6\ weeks}$, and 0.03 to cycle 3 C_{trough}.

Efficacy was assessed in the intention-to treat population. Efficacy findings were reported using descriptive statistics. The difference in ORR and the associated 95% CI were calculated using the stratified Miettinen and Nurminen's method[15] with strata weighting by sample size. The ORR ratio was calculated using the stratified Cochran–Mantel–Haenszel method. The Kaplan–Meier method was used to estimate PFS, OS, and DOR. A stratified Cox proportional hazards model with Efron's method of tie handling was used to calculate the hazard ratio (HR) and associated 95% CI between treatment arms. Stratification factors were applied to all stratified analyses.

Safety was assessed in all randomized participants who received at least one dose of study treatment. Safety findings were reported using descriptive statistics.

RESULTS

Participants and treatment

Of 640 participants screened for eligibility, 377 participants at 61 global sites were randomized between 2 March 2023 and 5 January 2024 to pembrolizumab s.c. plus chemotherapy ('pembrolizumab s.c. arm', $n = 251$) or pembrolizumab i.v. plus chemotherapy ('pembrolizumab i.v. arm', $n = 126$) arm (Figure 1; Supplementary Table S2, available at https://doi.org/10.1016/j.annonc.2025.03.012). Baseline demographics and clinical characteristics were balanced overall between treatment groups (Table 1). The median follow-up, defined as the time from randomization to the date of data cut-off, 12 July 2024, was 9.6 months (range 6.24-16.39 months). All 377 randomized participants received treatment. The median time on study treatment was 6.87 months (range 1 day-13.2 months) in the pembrolizumab s.c. arm and 6.21 months (range 1 day-15.9 months) in the pembrolizumab i.v. arm, and the median number of cycles of study treatment was 5 (range 1-10) and 5 (range 1-12), respectively. Most participants received carboplatin + pemetrexed chemotherapy (54.6% and 53.2% in the pembrolizumab s.c. and pembrolizumab i.v. arms, respectively), followed by carboplatin + paclitaxel (21.1% and 20.6%), carboplatin + nab-paclitaxel (12.4% and 13.5%), and cisplatin + pemetrexed (12.0% and 12.7%). The median injection time per administration of pembrolizumab s.c. was 2.0 min (range 1-12 min). The most common site of pembrolizumab s.c. injection was the abdominal wall.

Pharmacokinetics

The geometric means (95% CI) for cycle 1 $AUC_{0\text{-}6\ weeks}$ and cycle 3 C_{trough} were similar between treatment arms (Table 2). The GMR (96% CI) for cycle 1 $AUC_{0\text{-}6\ weeks}$ was 1.14 (1.06-1.22); $P < 0.0001$. The GMR (94% CI) for steady-state C_{trough} was 1.67 (1.52-1.84); $P < 0.0001$. Pembrolizumab AUC exposures after administration of pembrolizumab 790 mg s.c. q6w were largely within the range of those after administration of pembrolizumab 400 mg i.v. q6w.

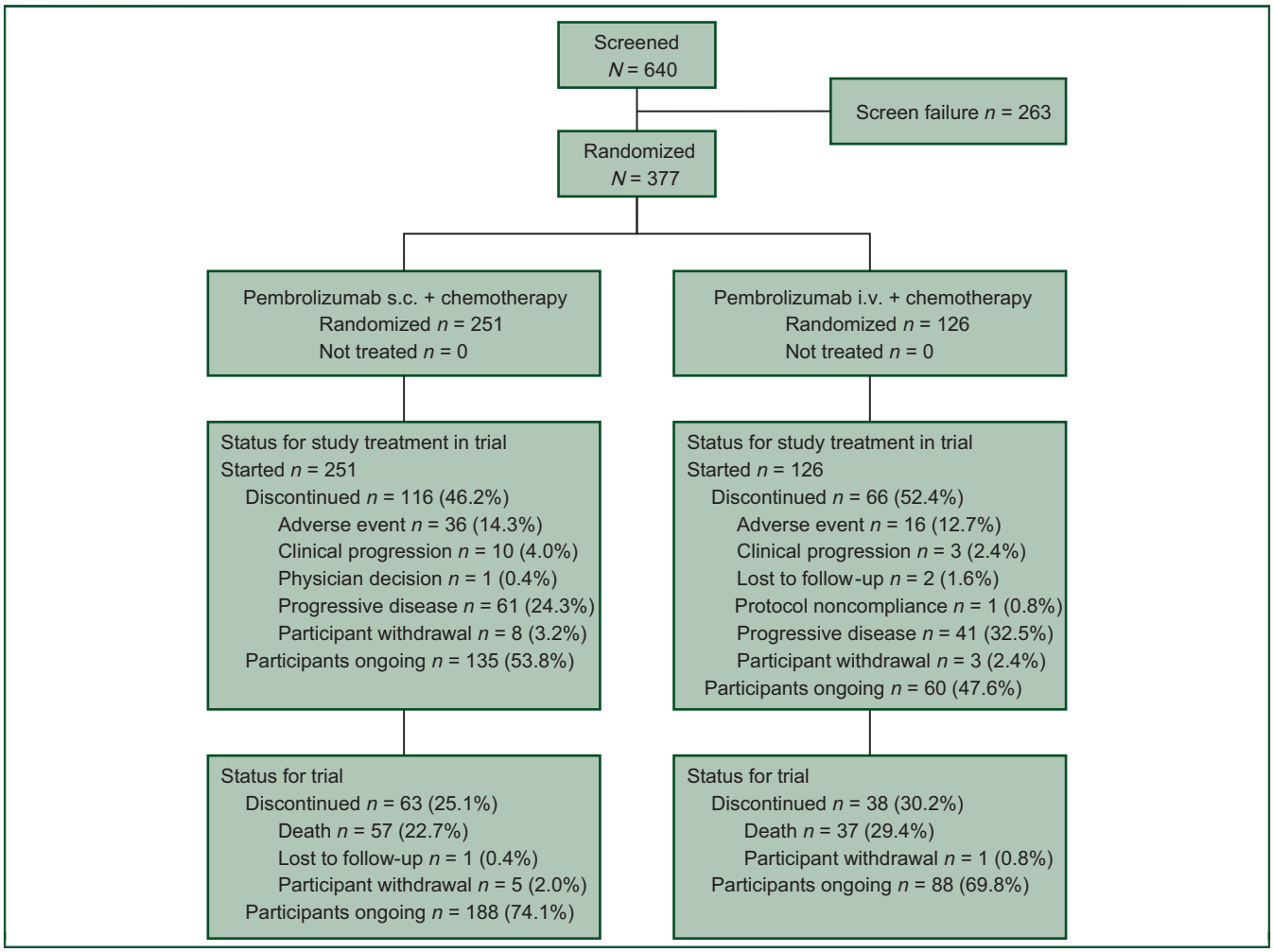


Figure 1. CONSORT diagram.
CONSORT, Consolidated Standards of Reporting Trials; i.v., intravenous; s.c., subcutaneous.

Pembrolizumab 790 mg s.c. q6w administration was associated with relatively higher cycle 1 C_{trough} and steady-state $AUC_{0\text{-}6\ weeks}$, and relatively lower C_{max} in both cycles 1 and 3 (Table 2). In addition, model-based C_{trough} for pembrolizumab 790 mg s.c. q6w was comparable to model-based C_{trough} for pembrolizumab 200 mg i.v. q3w in cycle 1 and at steady state. The results of observed assessments were consistent with the results of model-based assessments (data not shown).

Immunogenicity

The observed incidence of treatment-emergent ADAs among assessable participants was 1.4% ($n = 3$) in the pembrolizumab s.c. arm and 0.9% ($n = 1$) in the pembrolizumab i.v. arm (Supplementary Table S3, available at https://doi.org/10.1016/j.annonc.2025.03.012). One (0.5%) participant in the pembrolizumab s.c. arm who was treatment-emergent positive had neutralizing antibodies detected. For participants who were ADA positive, the pembrolizumab exposure was comparable to that for participants who were ADA negative and treated by the same route of administration. Treatment-emergent ADAs against berahyaluronidase alfa were detected in three (1.5%) participants with low titers (Supplementary Table S3, available at https://doi.org/10.1016/j.annonc.2025.03.012).

Efficacy

The ORR was 45.4% (95% CI 39.1% to 51.8%) in the pembrolizumab s.c. arm and 42.1% (95% CI 33.3% to 51.2%) in the pembrolizumab i.v. arm (ORR ratio 1.08, 95% CI 0.85-1.37) (Supplementary Table S4, available at https://doi.org/10.1016/j.annonc.2025.03.012). Outcomes were generally similar across subgroups (Figure 2). The median DOR was 9.1 months (range 6.9 months-not reached) in the pembrolizumab s.c. arm and 8.0 months (range 7.4 months-not reached) in the pembrolizumab i.v. arm (Supplementary Figure S1, available at https://doi.org/10.1016/j.annonc.2025.03.012). The best percentage change from baseline for target lesions is shown in Supplementary Figure S2, available at https://doi.org/10.1016/j.annonc.2025.03.012.

The median PFS was 8.1 months (95% CI 6.3-8.3 months) in the pembrolizumab s.c. arm and 7.8 months (95% CI 6.2-9.7 months) in the pembrolizumab i.v. arm (HR 1.05, 95% CI

Table 1. Participant baseline demographics and clinical characteristics

	Pembrolizumab s.c. + chemotherapy (*N* = 251)	Pembrolizumab i.v. + chemotherapy (*N* = 126)
Age, years		
Median (range)	65.0 (39-87)	66.0 (37-83)
<65, *n* (%)	119 (47.4)	57 (45.2)
≥65, *n* (%)	132 (52.6)	69 (54.8)
Sex, *n* (%)		
Male	182 (72.5)	86 (68.3)
Female	69 (27.5)	40 (31.7)
Race, *n* (%)		
American Indian or Alaska Native	2 (0.8)	4 (3.2)
Asian	74 (29.5)	36 (28.6)
Black or African American	5 (2.0)	5 (4.0)
Multiple	12 (4.8)	3 (2.4)
White	158 (62.9)	78 (61.9)
Geographic region, *n* (%)		
East Asia	73 (29.1)	36 (28.6)
North America/Western Europe/Australia/New Zealand	11 (4.4)	7 (5.6)
Rest of the world	167 (66.5)	83 (65.9)
PD-L1 status, *n* (%)		
TPS <50%	181 (72.1)	91 (72.2)
TPS <1%	101 (40.2)	57 (45.2)
TPS 1%-49%	80 (31.9)	34 (27.0)
TPS ≥50%	48 (19.1)	25 (19.8)
Unknown	22 (8.8)	10 (7.9)
ECOG performance status, *n* (%)		
0	89 (35.5)	42 (33.3)
1	162 (64.5)	84 (66.7)
Histology, *n* (%)		
Nonsquamous	167 (66.5)	83 (65.9)
Squamous	84 (33.5)	43 (34.1)
Overall stage, *n* (%)		
IVA	141 (56.2)	64 (50.8)
IVB	110 (43.8)	62 (49.2)
Metastasis stage, *n* (%)		
M1a	97 (38.6)	51 (40.5)
M1b	44 (17.5)	13 (10.3)
M1c	110 (43.8)	62 (49.2)
Brain metastases at baseline, *n* (%)	19 (7.6)	14 (11.1)
Liver metastases at baseline, *n* (%)	46 (18.3)	15 (11.9)
Smoking status,[a] *n* (%)		
Never	38 (15.1)	23 (18.3)
Former	142 (56.6)	62 (49.2)
Current	71 (28.3)	41 (32.5)
Prior adjuvant/neoadjuvant therapy, *n* (%)	5 (2.0)	6 (4.8)
Prior radiation, *n* (%)	37 (14.7)	24 (19.0)
Prior thoracic radiation, *n* (%)	10 (4.0)	8 (6.3)

ECOG, Eastern Cooperative Oncology Group; i.v., intravenous; PD-L1, programmed death-ligand 1; TPS, tumor proportion score, s.c., subcutaneous.
[a]Smokers were persons who reported smoking ≥100 cigarettes during their lifetime and who, at the time of screening, reported smoking every day or some days within the last year ("current") or had stopped smoking 1 year prior ("former"); never smokers were persons who reported smoking <100 cigarettes during their lifetime.

0.78-1.43) (Figure 3). The HRs across prespecified subgroups were generally consistent with that of the overall population (Supplementary Figure S3, available at https://doi.org/10.1016/j.annonc.2025.03.012).

The OS event rate was comparable between the pembrolizumab s.c. and pembrolizumab i.v. arms (24.3% versus 29.4%). As of data cut-off, the median OS has not been reached in either treatment arm (HR 0.81, 95% CI 0.53-1.22).

Safety

The overall safety profile was consistent between pembrolizumab s.c. + chemotherapy and pembrolizumab i.v. + chemotherapy. The incidences of treatment-related AEs, grade 3-5 treatment-related AEs, and serious treatment-related AEs were similar between treatment arms (Table 3; Supplementary Tables S5 and S6, available at https://doi.org/10.1016/j.annonc.2025.03.012). Treatment-related AEs led 8.4% of participants to discontinue pembrolizumab s.c. and 8.7% of participants to discontinue pembrolizumab i.v. In addition, 15.1% and 11.9% of participants in the pembrolizumab s.c. and i.v. arms, respectively, discontinued chemotherapy due to a treatment-related AE. In total, 38 participants in the study had an AE that resulted in death (Supplementary Table S7, available at https://doi.org/10.1016/j.annonc.2025.03.012), including 26 (10.4%) participants in the pembrolizumab s.c. + chemotherapy arm and 12 (9.5%) participants in the pembrolizumab i.v. + chemotherapy arm. Nine deaths (3.6%) in the pembrolizumab s.c. arm (three events of febrile neutropenia and one event each of neutropenic colitis, neutropenic sepsis, parotitis, pneumonia, septic shock, and pneumonitis) and three deaths (2.4%) in the pembrolizumab i.v. arm (two events of septic shock and one event of multiple organ dysfunction syndrome) were determined by the investigator to be related to treatment.

Similar proportions of participants in the two treatment arms experienced one or more serious AEs (Supplementary Table S7, available at https://doi.org/10.1016/j.annonc.2025.03.012), the most frequent of which included pneumonia, febrile neutropenia, thrombocytopenia, neutropenia, anemia, and pulmonary embolism.

The incidence, type, and severity of immune-mediated AEs reported for the pembrolizumab s.c. + chemotherapy arm were generally consistent with those reported for the pembrolizumab i.v. + chemotherapy arm (Supplementary Table S8, available at https://doi.org/10.1016/j.annonc.2025.03.012). Hypothyroidism was the most common immune-mediated AE in both treatment arms.

Injection-site reactions in the pembrolizumab s.c. arm were infrequent, reported in 6 (2.4%) participants, including injection-site reaction ($n = 2$) and one case each of injection-site erythema, hemorrhage, induration, and pain. All injection-site AEs were mild (grade 1) and nonserious. The median time to onset of the first injection-site AEs from the most recent dose administration was 2.0 days (range 1-2 days). The median duration of the first event of injection-site AE was 5.5 days (range 2-20 days). No participant discontinued treatment in the pembrolizumab s.c. arm due to injection-site AEs.

Health-related quality of life

Global health status/QoL, physical functioning, and role functioning as measured by the EORTC QLQ-C30 and overall health and QoL as measured by the EQ-5D-5L visual analogue scale score were generally stable between

Table 2. Analysis of pharmacokinetic measures

Dual primary endpoints	Pembrolizumab s.c. + chemotherapy	Pembrolizumab i.v. + chemotherapy
Cycle 1 $AUC_{0\text{-}6\ weeks}$ (μg·day/ml)	$N = 245$[a]	$N = 126$
GM (95% CI)	1633.24 (1555.23-1715.15)	1437.58 (1373.68-1504.46)
Geometric %CV	40.4	26.2
GMR (96% CI)[b]	1.14 (1.06-1.22); $P < 0.0001$[c]	
Cycle 3 model-based C_{trough} (μg/ml)	$N = 202$	$N = 101$
GM (95% CI)	39.23 (37.04-41.55)	23.49 (21.61-25.54)
Geometric %CV	43.3	44.2
GMR (94% CI)[b]	1.67 (1.52-1.84); $P < 0.0001$[d]	
Secondary PK endpoints (model based)	**Pembrolizumab 790 mg s.c. q6w**	**Pembrolizumab 400 mg i.v. q6w**
Cycle 1 C_{trough} (μg/ml)	$N = 245$[a]	$N = 126$
GM (%CV)	19.36 (52.2)	12.26 (50.8)
Difference[e] in GM, %	57.9	
Cycle 1 C_{max} (μg/ml)		
GM (%CV)	64.88 (44.0)	129.7 (20.8)
Difference[e] in GM, %	−50.0	
Cycle 3 C_{max} (μg/ml)	$N = 202$	$N = 101$
GM (%CV)	98.96 (36.8)	148 (21.6)
Difference[e] in GM, %	−33.1	
Cycle 3 $AUC_{0\text{-}6\ weeks}$ (μg·day/ml)		
GM (%CV)	2798 (35.5)	2122 (27.8)
Difference[e] in GM, %	31.9	
	Pembrolizumab 790 mg s.c. q6w	**Pembrolizumab 200 mg i.v. q3w**
Early cycle, week 6 (model based)		
C_{trough} (μg/ml)	$N = 245$[a]	$N = 126$
GM (%CV)	19.36 (52.2)	20.02 (35.8)
Difference[f] in GM, %	−3.3	
Steady state[g]		
C_{trough} (μg/ml)	$N = 202$	$N = 101$
GM (%CV)	39.23 (43.3)	34.51 (35.2)
Difference[f] in GM, %	13.7	

$AUC_{0\text{-}6\ weeks}$, area under the curve from week 0 to week 6; CI, confidence interval; C_{trough}, trough concentration; CV, coefficient variation; GM, geometric mean; GMR, geometric mean ratio; *N*, number of participants with evaluable PK data; N/A, not applicable; PK, pharmacokinetic; q3w, every 3 weeks; q6w, every 6 weeks.
[a]Six participants in the pembrolizumab s.c. arm were excluded from the PK modeling analysis, due to clinically meaningful protocol deviation ($n = 4$) and absence of cycle 1 samples for PK analysis ($n = 2$).
[b]Pembrolizumab s.c. + chemotherapy versus pembrolizumab i.v. + chemotherapy.
[c]The one-sided *P* value noninferiority boundary is 0.02 for the analysis of cycle 1 $AUC_{0\text{-}6\ weeks}$.
[d]The one-sided *P* value noninferiority boundary is 0.03 for the analysis of cycle 3 (steady-state) C_{trough}.
[e]Difference between model-based pembrolizumab s.c. 790 mg q6w and model-based pembrolizumab i.v. 400 mg q6w.
[f]Difference between model-based pembrolizumab s.c. 790 mg q6w and model-based pembrolizumab i.v. 200 mg q3w.
[g]Steady state is at cycle 3 for pembrolizumab 790 mg s.c. q6w and cycle 2 for pembrolizumab 200 mg i.v. q3w.

baseline and week 24 for each treatment arm with no difference between treatment arms (Supplementary Table S9, available at https://doi.org/10.1016/j.annonc.2025.03.012).

DISCUSSION

The phase III 3475A-D77 study demonstrated that pembrolizumab AUC exposure in cycle 1 and trough concentration at steady state were not inferior with pembrolizumab s.c. 790 mg q6w versus pembrolizumab i.v. 400 mg q6w, each administered in combination with platinum-doublet chemotherapy in participants with previously untreated, metastatic squamous and nonsquamous NSCLC. As expected, pembrolizumab exposures at steady state were higher with pembrolizumab s.c. versus i.v., given that the amount of pembrolizumab per s.c. administration was selected to ensure that the exposure range for efficacy is maintained. Higher exposures notwithstanding, the exposure level at steady state with pembrolizumab s.c. administration did not exceed the established clinical safety margin for pembrolizumab at doses up to 10 mg/kg q2w i.v.[16] Moreover, cycle 1 and steady-state peak concentrations of pembrolizumab in the pembrolizumab s.c. arm remained below those in the pembrolizumab i.v. arm.

Our study used a noninferiority design to evaluate the pharmacokinetics of pembrolizumab s.c. plus chemotherapy versus pembrolizumab i.v. plus chemotherapy, an approach that conforms to industry guidance and was employed to bridge the use of s.c. versus i.v. routes of administration for other cancer immunotherapies.[17-20] The IMscin001 study evaluated s.c. versus i.v. atezolizumab q3w for the treatment of locally advanced NSCLC or mNSCLC,[20] and the CheckMate 67T study compared s.c. nivolumab q4w with i.v. nivolumab q2w for the treatment of advanced or metastatic RCC.[19] Both studies met their co-primary pharmacokinetic endpoints and showed comparable efficacy and safety between the s.c. and i.v. arms. Notably, in both studies, immunogenicity was more pronounced with the s.c. drug than with the i.v. drug (IMscin001: 19.5% versus 13.9%; CheckMate 67T: 24.0% versus 6.9%), although neither study observed a meaningful clinical effect of ADA positivity. In our study, only limited formation of pembrolizumab ADAs was found (1.4% and

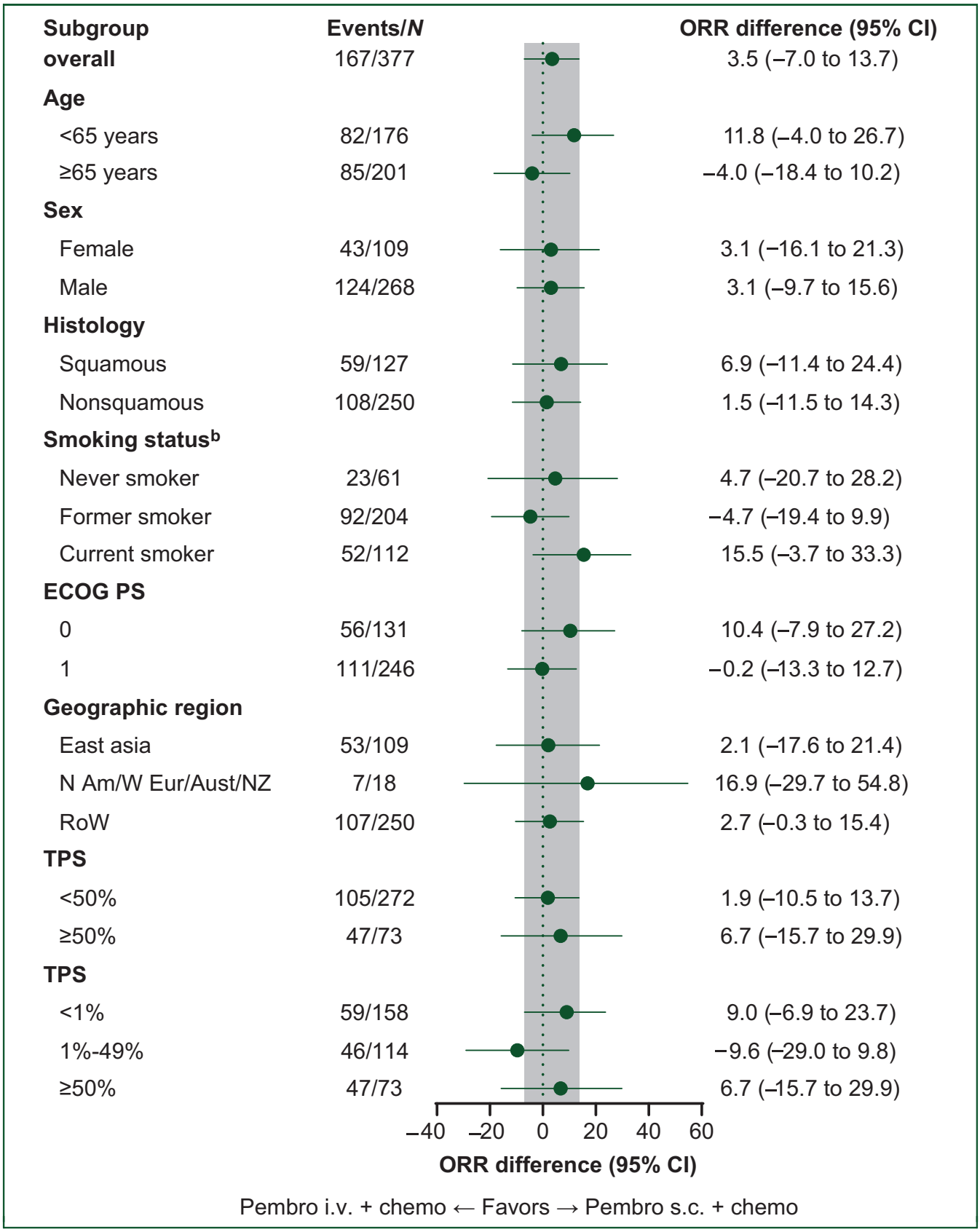


Figure 2. Forest plot of difference[a] in objective response rate based on blinded independent central review as per RECIST 1.1, by subgroup factors.
CI, confidence interval; ECOG, Eastern Cooperative Oncology Group; HR, hazard ratio; i.v., intravenous; N Am/W Eur/Aust/NZ, North America, western Europe, Australia, and New Zealand; PS, performance status; RoW, rest of the world; s.c., subcutaneous; TPS, tumor proportion score. [a]Difference between pembrolizumab s.c. + chemotherapy and pembrolizumab i.v. + chemotherapy. [b]Smokers were persons who reported smoking ≥100 cigarettes during their lifetime and who, at the time of screening, reported smoking every day or some days within the last year ("current") or had stopped smoking 1 year prior ("former"); never smokers were persons who reported smoking <100 cigarettes during their lifetime.

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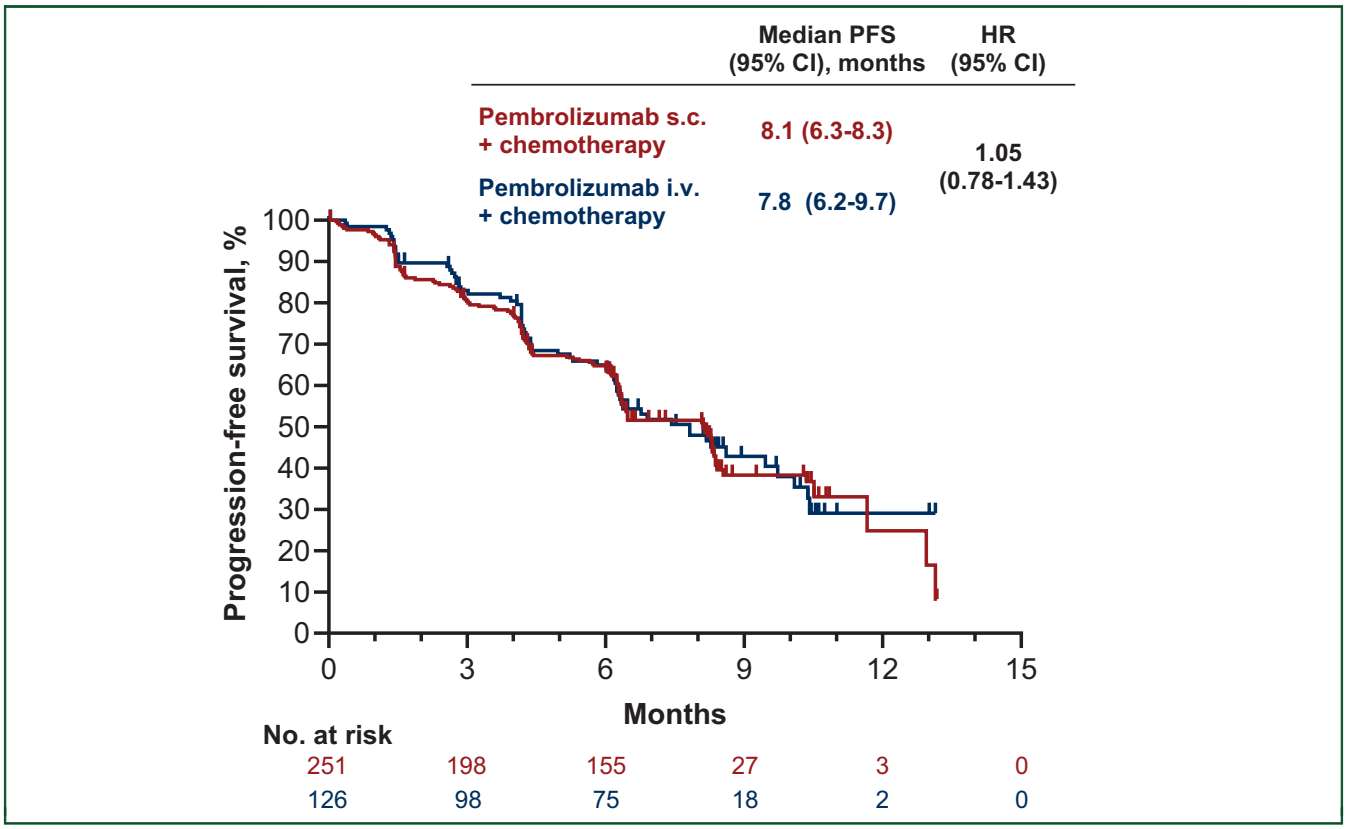


Figure 3. Kaplan–Meier estimates of PFS in the overall study population. PFS was based on blinded independent central review as per RECIST v1.1.
CI, confidence interval; HR, hazard ratio; i.v., intravenous; PFS, progression-free survival; s.c., subcutaneous.

Table 3. Summary of treatment-related adverse events

Participants[a], *n* (%)	Pembrolizumab s.c. + chemotherapy (*N* = 251)	Pembrolizumab i.v. + chemotherapy (*N* = 126)
With one or more adverse events	249 (99.2)	123 (97.6)
With treatment-related[b] adverse events	226 (90.0)	121 (96.0)
With grade 3-5 treatment-related[b] adverse events	118 (47.0)	60 (47.6)
With serious treatment-related[b] adverse events	53 (21.1)	25 (19.8)
Who discontinued any treatment due to a treatment-related[b] adverse event	44 (17.5)	19 (15.1)
Who discontinued pembrolizumab s.c. or i.v.	21 (8.4)	11 (8.7)
Who discontinued chemotherapy	38 (15.1)	15 (11.9)
Who died due to a treatment-related[b] adverse event	9 (3.6)[c]	3 (2.4)[d]

i.v., intravenous; s.c., subcutaneous.
[a]Participants are counted a single time for each applicable row and column. Nonserious adverse events up to 30 days after the last dose and serious adverse events up to 90 days after the last dose are included.
[b]Determined by the investigator to be related to treatment.
[c]Includes febrile neutropenia ($n = 3$), neutropenic colitis ($n = 1$), neutropenic sepsis ($n = 1$), parotitis ($n = 1$), pneumonia ($n = 1$), septic shock ($n = 1$), and pneumonitis ($n = 1$).
[d]Includes septic shock ($n = 2$) and multiple organ dysfunction syndrome ($n = 1$).

0.9% in the pembrolizumab s.c. and i.v. arms, respectively), with no meaningful clinical impact of ADAs on exposure. The detection of pembrolizumab ADAs (1.4%) and neutralizing antibodies (0.5%) among participants in the pembrolizumab s.c. arm was comparable to the results of an immunogenicity analysis of pembrolizumab i.v. ($N = 2000$ assessable participants), in which ADAs were detected in 1.8% of participants and neutralizing antibodies in 0.5%.[21]

Consistent with the noninferiority of pharmacokinetic exposures of pembrolizumab s.c. and i.v., efficacy was similar between treatment arms. The 3475A-D77 was not, however, designed as a noninferiority study with respect to the secondary efficacy endpoints. Moreover, ORR, median PFS, and median DOR reported for the two treatment arms were consistent with those reported in the KEYNOTE-189 and KEYNOTE-407 trials of pembrolizumab i.v. + chemotherapy for the treatment of mNSCLC.[22,23] As of this analysis, the median duration of follow-up in the study is shorter than those of the KEYNOTE-189 and KEYNOTE-407 studies. Therefore, the OS data are considered immature. In addition, the subgroup analysis results for ORR and PFS remained consistent across all analyzed groups.

The safety profiles of pembrolizumab s.c. and pembrolizumab i.v. were generally consistent, including the incidence rates of AEs and rates of discontinuation of pembrolizumab s.c. and pembrolizumab i.v.. There were no unexpected safety findings with pembrolizumab s.c.

treatment. Local injection-site AEs with pembrolizumab s.c. administration were infrequent, mild, and nonserious.

Shorter administration time has been touted as an advantage of the s.c. route. The median administration time for pembrolizumab s.c. injection in this study was 2 min, which was shorter than the typical infusion time of ∼30 min for pembrolizumab i.v., as well as the median of 5 min for s.c. nivolumab administration reported in the CheckMate 67T study.[19] The IMscin001 study did not report a median administration time, although <10 min was the target administration time for s.c. atezolizumab.[20] A time and motion study suggested that pembrolizumab s.c. can improve administration workflow by reducing active time for providers and chair time for patients.[24]

Treatment-naive mNSCLC was chosen as the indication to evaluate in this pharmacokinetic noninferiority study because lung cancer remains the most common malignancy worldwide.[25] Establishing the pharmacokinetic noninferiority of pembrolizumab s.c. to i.v., both given with chemotherapy, as well as comparable efficacy and safety in NSCLC, may allow bridging to other approved pembrolizumab indications. Further, the pembrolizumab s.c. q6w regimen may allow for increased flexibility, especially for patients who would receive pembrolizumab in the early-stage setting, or in combination with chemotherapies or other agents of varying dosing schedules.

In summary, the 3475A-D77 study showed that overall exposure and trough concentration of pembrolizumab s.c. were noninferior to those of pembrolizumab i.v., as well as similar efficacy and consistent safety profiles between the pembrolizumab s.c. and i.v. treatment arms. The immunogenicity of pembrolizumab s.c. was low, and local injection-site AEs were infrequent, mild, and nonserious. The injection administration time was notably short. These results demonstrate that pembrolizumab s.c. is a treatment option for all indications where pembrolizumab i.v. can be used, with the potential to improve the patient experience and reduce time demands in the clinic, which may streamline treatment center workflows and reduce health care resource utilization.

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DATA SHARING

Merck Sharp & Dohme LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA (MSD) is committed to providing qualified scientific researchers access to anonymized data and clinical study reports from the company's clinical trials for the purpose of conducting legitimate scientific research. MSD is also obligated to protect the rights and privacy of trial participants and, as such, has a procedure in place for evaluating and fulfilling requests for sharing company clinical trial data with qualified external scientific researchers. The MSD data-sharing website (available at: https://externaldatasharing-msd.com/) outlines the process and requirements for submitting a data request. Applications will be promptly assessed for completeness and policy compliance. Feasible requests will be reviewed by a committee of MSD subject matter experts to assess the scientific validity of the request and the qualifications of the requestors. In line with data privacy legislation, submitters of approved requests must enter into a standard data-sharing agreement with MSD before data access is granted. Data will be made available for request after product approval in the United States and European Union or after product development is discontinued. There are circumstances that may prevent MSD from sharing requested data, including country or region-specific regulations. If the request is declined, it will be communicated to the investigator. Access to genetic or exploratory biomarker data requires a detailed, hypothesis-driven statistical analysis plan that is collaboratively developed by the requestor and MSD subject matter experts; after approval of the statistical analysis plan and execution of a data-sharing agreement, MSD will either carry out the proposed analyses and share the results with the requestor or will construct biomarker covariates and add them to a file with clinical data that is uploaded to an analysis portal so that the requestor can carry out the proposed analyses.

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